# Fair by design

## Optimization · Level 4 · Lesson 2



# Today we will

- Score a plan by its total, its worst-off and its gap.
- Find the plan that makes the least happy person as happy as possible.
- Measure the price of fairness and agree a rule together.

# Remember

- The total can hide who pays; look at the worst-off too.
- Fairness goes into the model as a floor, a goal or a limit on the gap.
- Agree the rule before anyone sees the result, then publish it.

# Who carries the cost?

1. Warm-up (5 min): Is the biggest total always fair?
2. Teach (8 min): Total, worst-off and gap.
3. Solve (12 min): Two groups, two goals, the same table.
4. Compare (8 min): What did fairness cost?
5. Agree (7 min): Which rule would the class choose?

**Use real class jobs and let pupils give their own scores if they like; keep the table to three pupils and three jobs so every plan can be checked by hand.**

# Let’s discuss

1. When is a small price of fairness worth paying, and when is a large one?
2. Who should choose the fairness rule for our school’s duties?
3. How can fairness work over several terms, not just one?

# Exit ticket

- Write one fairness rule for a school decision, and say whether it is a floor, a goal or a limit on the gap.